

Yazhuo Zhang

☎ (+41) 076-229-3975 | ✉ z.yazhuo@gmail.com | 🏠 <http://yazhuozhang.com> | 🌐 yazhuo

Academic Position

Post-doctoral researcher in EASL Lab, ETH Zurich
Computer Science Department, Advisor: Ana Klimovic

Zurich, Switzerland
July.2024 - Present

Interests

Improve the performance and resource efficiency of large-scale distributed systems

Education

Ph.D. in Computer Science, Emory University
Computer Science Department, Advisor: Ymir Vigfusson

Atlanta, USA
Aug.2019 - May.2024

M.S. in Software Engineering, South China University of Technology
Software Engineering Department, Advisor: Deyou Tang

Guangzhou, China
Sep.2016 - Jun.2019

B.S. in Software Engineering, South China University of Technology
Software Engineering Department, Advisor: Deyou Tang

Guangzhou, China
Sep.2012 - Jun.2016

Publications

- Yazhuo Zhang**, Jinqing Cai, Avani Wildani, Ana Klimovic. [Rethinking Web Cache Design for the AI Era](#). *The 16th edition of the annual ACM Symposium on Cloud Computing (SoCC)*, 2025
- Tom Kuchler, Pinghe Li, **Yazhuo Zhang**, Lazar Cvetković, Boris Goranov, Tobias Stocker, Leon Thomm, Simone Kalbermatter, Tim Notter, Andrea Lattuada, Ana Klimovic. [Unlocking True Elasticity for the Cloud-Native Era with Dandelion](#). *The 31th ACM Symposium on Operating Systems Principles (SOSP)*, 2025.
- Yazhuo Zhang**, Juncheng Yang, Yao Yue, Ymir Vigfusson, Rashmi Vinayak. [SIEVE: Simple and Efficient Eviction Policy for Turn-key Web Cache Replacement](#). *The 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2024. **Community Award**.
- Yazhuo Zhang**, Rebecca Isaacs, Yao Yue, Juncheng Yang, Lei Zhang, Ymir Vigfusson. [LatenSeer: Causal Modeling of End-to-End Latency Distribution by Harnessing Distributed Tracing](#). *14th ACM Symposium on Cloud Computing (SoCC)*, 2023.
- Juncheng Yang, **Yazhuo Zhang**, Ziyue Qiu, Yao Yue, Rashmi Vinayak. [FIFO queues are all you need for cache eviction](#). *The 29th ACM Symposium on Operating Systems Principles (SOSP)*, 2023.
- Juncheng Yang, Ziyue Qiu, **Yazhuo Zhang**, Yao Yue, Rashmi Vinayak. [FIFO can be Better than LRU: the Power of Lazy Promotion and Quick Demotion](#). *The 19th Workshop on Hot Topics in Operating Systems (HotOS)*, 2023.

Industry Research Experience

Research Intern @ Magnition

CDN Performance What-if Analysis

June. 2023 - Present

Manager/Mentor: Irfan Ahmad

- Conducted a thorough survey of the existing CDN architecture and performance requirements through reviewing academic literature and interviewing CDN industry experts.
- Designed and implemented a versatile CDN simulator utilizing the Lingua Franca language. This simulator facilitates explorations of various parameter spaces, including but not limited to resource allocation and eviction policies.
- Currently engaged in the rigorous assessment of the CDN simulator's performance and applicability in real-world scenarios, aiming to optimize its effectiveness and utility.

Research Intern @ Akamai

MetroCache Routing Performance Analysis

May 2022 - Aug 2022

Manager: Anna Blasiak, Mentor: Scott Roche

- Collected traffic data on test machines with the geo-choice feature enabled, providing geo-partition and geo-choice results for each ECOR.
- Designed and conduct experiments focused on determining the impact of various parameter settings on geo-choice results, leading to a deep understanding of geo-choice performance and its correlation with parameter adjustments.
- Identified optimal parameters' range for geo-choice feature. Additionally, uncovered potential capacity deficiencies in specific regions.
- Devised an interactive tool for for monitoring geo-choice feature's performance, thus enhancing the company's capacity to promptly identify and rectify performance-related issues.

Researcher @ Twitter

What-if Analysis for Hypothetical Service Migration

Feb 2021 - May 2022

Manager: Yao Yue, Mentor: Rebecca Isaacs

- Enhanced Google Cloud Dataflow for the construction of service dependency trees, capable of querying tens of millions of Zipkin traces and storing crucial details of sibling spans.
- Devised a latency-critical path model, LatenSeer, which can estimate end-to-end latency distributions under hypothetical latency changes in internal services, further reporting each service's latency slack.
- Evaluated LatenSeer using prototypes and production traces. LatenSeer predicts end-to-end latency distribution within 5.35% error (D-Statistic).
- Helped investigate cache cluster behavior during failover periods and clock skew distributions among production machines.

Awards and Honors

2024	NSDI'24 diversity grant
2023	OSDI'23 diversity grant
2023	HotOS'23 travel grant
2022	OSDI'22 diversity grant
2021	OSDI'20 student grant
2018	Outstanding cadres of Student Association
2016	Excellence of Commercial Promise of Intel Cup Parallel Application Challenge

Teaching Experience

2026	Cloud Computing Architecture (Teaching Assistant, Guest Lecturer), <i>ETH Zurich</i>
2025	System Programming and Computer Architecture (Teaching Assistant), <i>ETH Zurich</i>
2025	Cloud Computing Architecture (Teaching Assistant, Guest Lecturer), <i>ETH Zurich</i>
2021	Analysis of Algorithms (Teaching Assistant), <i>Emory University</i>
2020	System programming (Teaching Assistant), <i>Emory University</i>
2020	Introduction to Computer Science (Teaching Assistant), <i>Emory University</i>
2018	Fundamentals of Computers (Teaching Assistant), <i>SCUT</i>
2017	Database (Teaching Assistant), <i>SCUT</i>
2017	Parallel Algorithms and Programming (Lab Instructor), <i>SCUT</i>

Services

2025	Reviewer: TOS
2024	Artifact Evaluation Committee: ATC’24
2024	Artifact Evaluation Committee: OSDI’24
2024	External Reviewer: EuroSys’24
2022	Volunteer: LADIS’22
2021	External Reviewer: ATC’21
2020	External Reviewer: HotCloud’20